Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2024-25 (ODD)	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	22	Name of Student	Sanskruti Kharapkar

Practical Number	6
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**To design and develop a Shopping Cart Web Application using HTML, CSS, and JavaScript that allows users to view products, add or remove items, modify quantities, and calculate the total bill dynamically.**
Problem Definition	**To create a website where users can view products, add or remove items, change quantities, and calculate the final bill automatically.**
Theory (100 words)	**HTML is used to create the structure of the shopping cart application, while CSS provides styling and a responsive layout. JavaScript adds interactivity and manages product and cart data using arrays of objects. The `map()` method displays products, `find()` searches for a product, `push()` adds products to the cart, and `filter()` removes selected items. The `reduce()` method calculates the subtotal based on product prices and quantities. Event handling allows users to update quantities and remove items. DOM manipulation updates the cart count and bill summary instantly without reloading the webpage. Thus, the application demonstrates dynamic web development.**

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Procedure and Execution (100 Words)	Step for Implementation: • Create a project folder named `ShopEase`. • Create `index.html`, `style.css`, and `script.js`. • Design the webpage structure using HTML. • Apply styling using CSS. • Create an array of product objects in JavaScript. • Implement Add to Cart, quantity control, and Remove functionality. • Calculate the bill using `reduce()` and update the webpage dynamically. • Run the website in a browser and test all operations.

Code:

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>ShopEase | Online Shopping</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js" defer></script>
</head>
<body>
  <header class="site-header">
    <a class="brand" href="#top" aria-label="ShopEase
home"><span class="brand-mark">S</span> ShopEase</a>
    <p class="tagline">Find your everyday favourites</p>
    <a class="cart-link" href="#cart-section"
aria-label="Jump to your cart">
      <span aria-hidden="true">🛒</span> Your Cart <span
class="cart-count" id="cart-count">0</span>
    </a>
```

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```
  </header>

  <main id="top" class="page-wrap">
    <section class="intro">
      <p class="eyebrow">SHOP THE COLLECTION</p>
      <h1>Explore Items</h1>
      <p class="intro-copy">Useful picks, everyday
essentials, and little upgrades for you.</p>
    </section>

    <section aria-label="Available products">
      <div id="product-list" class="product-grid"
aria-live="polite"></div>
    </section>

    <section id="cart-section" class="cart-section">
      <div class="section-heading">
        <div>
          <p class="eyebrow">YOUR SELECTION</p>
          <h2>Your Shopping Cart <span
class="muted-count" id="cart-item-count">(0
items)</span></h2>
        </div>
        <button id="clear-cart" class="text-button"
type="button" hidden>Clear cart</button>
      </div>
      <div id="cart-list" class="cart-list"
aria-live="polite"></div>
    </section>

    <section id="bill-section" class="bill-section">
      <div class="bill-card">
        <div class="bill-title-row">
          <div>
            <p class="eyebrow">ORDER TOTAL</p>
            <h2>Bill Summary</h2>
          </div>
```

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```
          <span class="bill-icon"
aria-hidden="true">🧾</span>
        </div>
        <div
class="bill-line"><span>Subtotal</span><strong
id="subtotal">₹0</strong></div>
        <div class="bill-line"><span>Delivery
charge</span><strong id="delivery">₹0</strong></div>
        <div class="bill-line
discount-line"><span>Discount</span><strong
id="discount">-₹0</strong></div>
        <div class="bill-line total-line"><span>Total
amount</span><strong id="total-price">₹0</strong></div>
        <p class="delivery-note" id="delivery-note">Add
items to see your order total.</p>
        <button id="checkout" class="checkout-button"
type="button">Proceed to Checkout <span
aria-hidden="true">→</span></button>
        <p id="message" class="checkout-message"
role="status" aria-live="polite"></p>
        <p class="secure-note"><span
aria-hidden="true">🔒</span> Demo checkout · No payment
will be collected</p>
      </div>
    </section>
  </main>

  <footer>ShopEase · Student JavaScript Practical
Project</footer>
</body>
</html>
```

JAVASCRIPT

```
// ShopEase demo shopping cart: products are sample items
and checkout is simulated.
const products = [
  {
```

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```
    id: 1,
    name: "Wireless Headphones",
    category: "Electronics",
    price: 1299,
    rating: "4.4",
    reviews: 128,
    badge: "Bestseller",
    image:
"https://images.unsplash.com/photo-1505740420928-5e560c06
d30e?auto=format&fit=crop&w=700&q=80"
  },
  {
    id: 2,
    name: "Classic Wrist Watch",
    category: "Accessories",
    price: 899,
    rating: "4.2",
    reviews: 86,
    badge: "Popular",
    image:
"https://images.unsplash.com/photo-1523275335684-37898b6b
af30?auto=format&fit=crop&w=700&q=80"
  },
  {
    id: 3,
    name: "Everyday Sneakers",
    category: "Footwear",
    price: 1599,
    rating: "4.5",
    reviews: 204,
    badge: "Top rated",
    image:
"https://images.unsplash.com/photo-1542291026-7eec264c27f
f?auto=format&fit=crop&w=700&q=80"
  },
  {
    id: 4,
```

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```
    name: "Travel Backpack",
    category: "Bags",
    price: 799,
    rating: "4.3",
    reviews: 67,
    badge: "Everyday pick",
    image:
"https://images.unsplash.com/photo-1553062407-98eeb64c6a62?auto=format&fit=crop&w=700&q=80"
  },
  {
    id: 5,
    name: "Classic Sunglasses",
    category: "Accessories",
    price: 499,
    rating: "4.1",
    reviews: 53,
    badge: "Deal",
    image:
"https://images.unsplash.com/photo-1511499767150-a48a237f0083?auto=format&fit=crop&w=700&q=80"
  },
  {
    id: 6,
    name: "Reusable Water Bottle",
    category: "Lifestyle",
    price: 299,
    rating: "4.6",
    reviews: 145,
    badge: "Value pick",
    image:
"https://images.unsplash.com/photo-1602143407151-7111542de6e8?auto=format&fit=crop&w=700&q=80"
  }
];
```

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```
// Each cart entry contains a product's fields and a
quantity.
let cart = [];
const rupees = amount =>
`₹${amount.toLocaleString("en-IN")}`;

const productList =
document.getElementById("product-list");
const cartList = document.getElementById("cart-list");
const cartCount = document.getElementById("cart-count");
const cartItemCount =
document.getElementById("cart-item-count");
const clearCartButton =
document.getElementById("clear-cart");
const checkoutMessage =
document.getElementById("message");

function displayProducts() {
  productList.innerHTML = products.map(product => `
    <article class="product-card">
      <div class="product-image-wrap">
        <img src="${product.image}" alt="${product.name}"
loading="lazy">
        <span
class="product-badge">${product.badge}</span>
      </div>
      <div class="product-info">
        <p
class="product-category">${product.category}</p>
        <h3 class="product-name">${product.name}</h3>
        <p class="rating">★
${product.rating}<span>(${product.reviews}
reviews)</span></p>
        <div class="product-bottom">
          <span
class="product-price">${rupees(product.price)}</span>
```

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```
          <button class="add-button" type="button"
data-add="${product.id}">+ Add to Cart</button>
        </div>
      </div>
    </article>
  `).join("");
}

function addToCart(id) {
  const product = products.find(item => item.id === id);
  if (!product) return;

  const existingItem = cart.find(item => item.id === id);
  if (existingItem) {
    existingItem.quantity += 1;
  } else {
    cart.push({ ...product, quantity: 1 });
  }
  checkoutMessage.textContent = "";
  displayCart();
}

function changeQuantity(id, change) {
  const item = cart.find(product => product.id === id);
  if (!item) return;

  item.quantity += change;
  if (item.quantity <= 0) {
    removeFromCart(id);
    return;
  }
  checkoutMessage.textContent = "";
  displayCart();
}

function removeFromCart(id) {
  cart = cart.filter(item => item.id !== id);
```

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```
  checkoutMessage.textContent = "";
  displayCart();
}

function clearCart() {
  cart = [];
  checkoutMessage.textContent = "";
  displayCart();
}

function calculateBill() {
  const subtotal = cart.reduce((sum, item) => sum +
item.price * item.quantity, 0);
  // Demo rules: free delivery at ₹999 and 10% off at
₹1,500.
  const delivery = subtotal === 0 || subtotal >= 999 ? 0
: 50;
  const discount = subtotal >= 1500 ? Math.round(subtotal
* 0.10) : 0;
  const total = subtotal + delivery - discount;

  document.getElementById("subtotal").textContent =
rupees(subtotal);
  document.getElementById("delivery").textContent =
subtotal === 0 ? rupees(0) : delivery === 0 ? "FREE" :
rupees(delivery);
  document.getElementById("discount").textContent =
`-${rupees(discount)}`;
  document.getElementById("total-price").textContent =
rupees(total);

  const note = document.getElementById("delivery-note");
  if (subtotal === 0) {
    note.textContent = "Add items to see your order
total.";
  } else if (subtotal < 999) {
```

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```
    note.textContent = `Add ${rupees(999 - subtotal)}
more to unlock free delivery.`;
  } else if (subtotal < 1500) {
    note.textContent = "Great! Your delivery is free.";
  } else {
    note.textContent = "Great! Free delivery and 10%
discount applied.";
  }
}

function displayCart() {
  const totalQuantity = cart.reduce((sum, item) => sum +
item.quantity, 0);
  cartCount.textContent = totalQuantity;
  cartItemCount.textContent = `(${totalQuantity}
${totalQuantity === 1 ? "item" : "items"})`;
  clearCartButton.hidden = cart.length === 0;

  if (cart.length === 0) {
    cartList.innerHTML = `
      <div class="empty-cart">
        <span class="empty-icon"
aria-hidden="true">🛍️</span>
        <strong>Your cart is empty</strong>
        <span>Explore the items above and add something
you like.</span>
      </div>`;
  } else {
    cartList.innerHTML = cart.map(item => `
      <article class="cart-item">
        <img class="cart-thumb" src="${item.image}"
alt="${item.name}">
        <div class="cart-product-info">
          <p class="cart-product-name">${item.name}</p>
          <p
class="cart-unit-price">${rupees(item.price)} each</p>
```

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```
          <button class="remove-button" type="button"
data-remove="${item.id}">Remove</button>
        </div>
        <div class="cart-item-actions">
          <div class="qty-control" aria-label="Quantity
for ${item.name}">
            <button class="qty-button" type="button"
data-change="-1" data-id="${item.id}"
aria-label="Decrease ${item.name} quantity">-</button>
            <span
class="qty-number">${item.quantity}</span>
            <button class="qty-button" type="button"
data-change="1" data-id="${item.id}" aria-label="Increase
${item.name} quantity">+</button>
          </div>
          <div
class="cart-item-total">${rupees(item.price *
item.quantity)}</div>
        </div>
      </article>
    `).join("");
  }

  calculateBill();
}

// Event delegation keeps all dynamically rendered
buttons working.
productList.addEventListener("click", event => {
  const button = event.target.closest("[data-add]");
  if (button) addToCart(Number(button.dataset.add));
});

cartList.addEventListener("click", event => {
  const removeButton =
event.target.closest("[data-remove]");
  if (removeButton) {
```

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```
    removeFromCart(Number(removeButton.dataset.remove));
    return;
  }

  const quantityButton =
event.target.closest("[data-change]");
  if (quantityButton) {
    changeQuantity(Number(quantityButton.dataset.id),
Number(quantityButton.dataset.change));
  }
});

clearCartButton.addEventListener("click", clearCart);

document.getElementById("checkout").addEventListener("cli
ck", () => {
  if (cart.length === 0) {
    checkoutMessage.textContent = "Your cart is empty.
Add a product before checkout.";
    return;
  }
  checkoutMessage.textContent = "Your demo order summary
is ready. No real order or payment has been placed.";
});

displayProducts();
displayCart();
```

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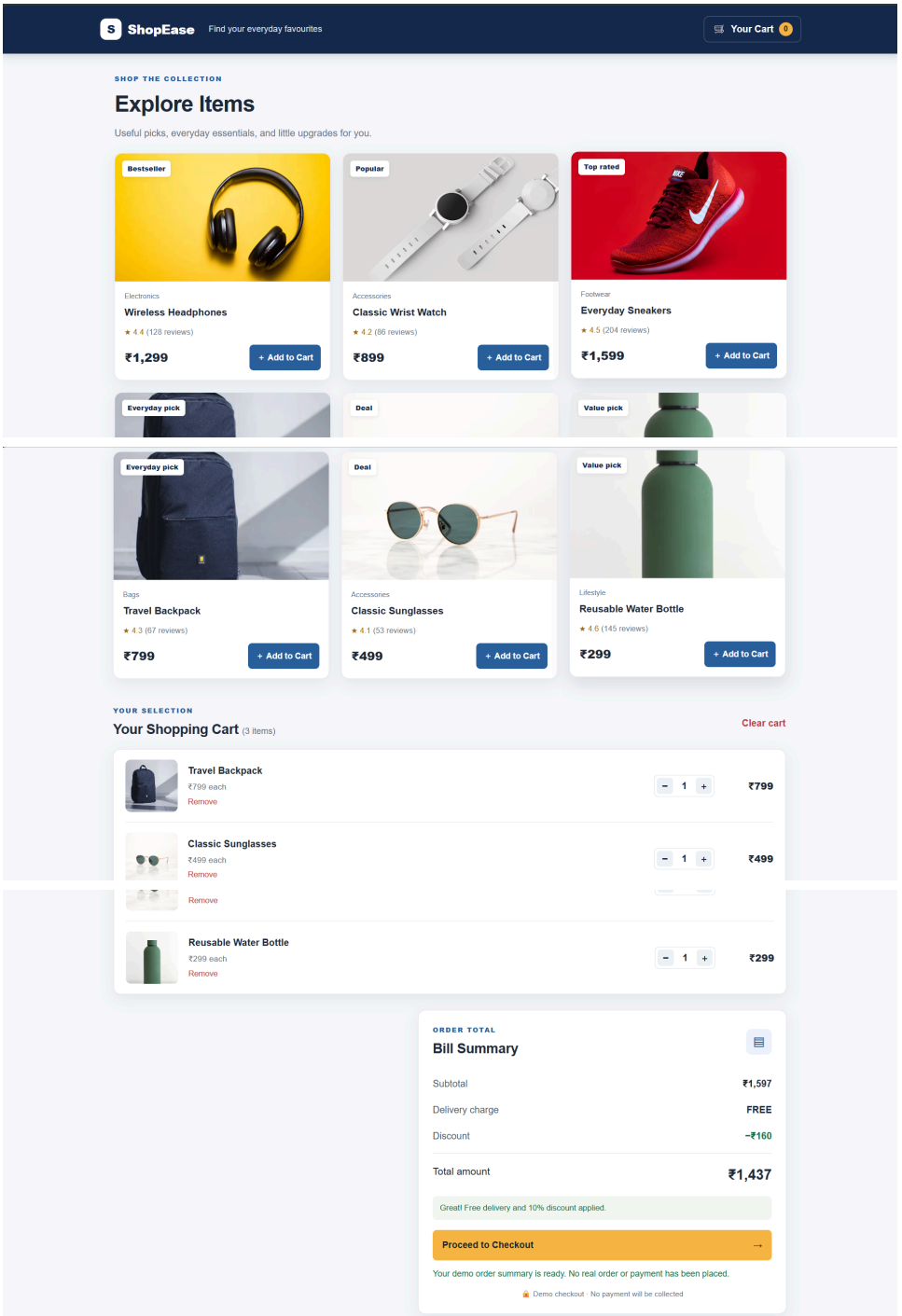
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Output:



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Output Analysis	**The application successfully demonstrates dynamic shopping cart management. The `map()` method renders the product cards and cart items, `find()` locates products, `push()` adds new cart entries, and `filter()` removes selected products. The `reduce()` method calculates the subtotal using the price and quantity of each item. Delivery charges are ₹50 for a non-empty cart below ₹999 and free for orders of ₹999 or more. A 10% discount is applied when the subtotal reaches ₹1,500. The final amount is recalculated whenever the cart changes. Checkout is implemented for demonstration purposes only.**
Link of student Github profile where lab assignment has been uploaded	**Sanskruti2007-code/web-Technology-Practicals**
Conclusion	**The ShopEase Shopping Cart Web Application demonstrates the integration of HTML, CSS, and JavaScript to create an interactive shopping website. It provides product display, cart management, quantity control, and automatic bill calculation. The practical helped in understanding arrays, objects, higher-order functions, event handling, and DOM manipulation. The project provides hands-on experience in developing a user-friendly web application similar to basic e-commerce websites.**
Plag Report (Similarity index < 12%)	**10%**
Date	**28/09/2026**